

DynDep: Dynamical Topology-Based Multimodal Depression Detection

Research Report

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Abstract

Existing multimodal depression detection systems primarily treat depression assessment as a static classification problem. Audio, facial expressions, gaze, head pose, and behavioral signals are fused to predict a binary depression label. While effective, such approaches largely ignore the temporal evolution of behavioral states throughout an interview.

In this work, we propose **DynDep**, a dynamical topology-based framework for multimodal depression detection. Instead of viewing depression as a static collection of features, we model an interview as a latent behavioral trajectory evolving over time. A multimodal encoder transforms sequential observations into latent states, producing a trajectory in behavioral state space.

We introduce three complementary classes of biomarkers. First, recurrence-based biomarkers quantify the frequency with which behavioral states revisit previous regions of latent space. Second, topological biomarkers derived from persistent homology characterize trajectory fragmentation, connectivity, cyclicity, and structural complexity. Third, dynamical stability biomarkers measure trajectory smoothness and temporal consistency.

These biomarkers are integrated with a classification network through a topology-aware learning objective that jointly optimizes predictive performance and trajectory regularity. The proposed framework provides both improved interpretability and a fundamentally different representation of depression as a dynamical process rather than a static classification task.

We further demonstrate that the proposed framework can operate as a plug-in extension to existing multimodal architectures, including transformer-based systems such as GATA-Dep. Our work establishes a new direction for depression assessment through latent behavioral dynamics and topological analysis.

1 Introduction

Depression is inherently a temporal disorder. Clinical manifestations emerge through gradual changes in speech, facial expressions, gaze behavior, and psychomotor activity. However, most machine learning approaches for depression detection focus on static feature representations and classification performance.

This creates a mismatch between computational models and clinical reality. While depression unfolds as a behavioral process, existing systems largely ignore the geometry and evolution of behavioral states during an interview.

We hypothesize that the trajectory itself contains diagnostic information. Specifically, healthy and depressed participants may exhibit distinct dynamical structures in latent behavioral space even when average feature statistics appear similar.

To investigate this hypothesis, we introduce **DynDep**, a framework that combines multimodal representation learning with tools from nonlinear dynamical systems and topological data analysis.

Key Contributions

- A trajectory-based formulation of depression detection.
- Recurrence-based behavioral biomarkers.
- Persistent homology biomarkers for latent behavioral trajectories.
- Topology-aware learning objectives for multimodal depression detection.
- An interpretable framework compatible with existing multimodal architectures.

2 Methodology

2.1 Latent Behavioral Trajectory

Given multimodal observations over T temporal windows, a multimodal encoder generates latent states:

$$z_1, z_2, \dots, z_T \tag{1}$$

forming a continuous trajectory in the latent space.

2.2 Recurrence Dynamics

A recurrence matrix is constructed from pairwise latent-state distances. From this matrix, we compute:

- **Recurrence Rate:** The density of recurrence points in the matrix.
- **Determinism:** The predictability of the system based on diagonal lines.
- **Laminarity:** The occurrence of laminar states based on vertical lines.
- **Recurrence Entropy:** The complexity of the deterministic behavior.

These metrics quantify behavioral consistency and state revisitation patterns.

2.3 Persistent Homology

Persistent homology is applied directly to the latent trajectory to extract structural invariants. Topological biomarkers include:

- **Betti-0** (β_0): Represents trajectory fragmentation.
- **Betti-1** (β_1): Measures cyclic behavior and looping dynamics.
- **Persistence Entropy:** Quantifies the distribution of topological feature lifetimes.
- **Average Persistence Lifetime:** Measures the robustness of the discovered topological structures.

These features capture global trajectory structure beyond conventional statistical moments.

2.4 Dynamical Stability

Temporal smoothness is measured through trajectory velocity and acceleration statistics. A dedicated smoothness regularizer encourages realistic, non-erratic behavioral evolution across time steps.

2.5 Topology-Aware Learning

The final composite optimization objective is defined as:

$$L = L_{\text{cls}} + \lambda_1 L_{\text{smooth}} + \lambda_2 L_{\text{topo}} \quad (2)$$

where L_{cls} denotes the standard classification loss, L_{smooth} promotes temporal consistency, and L_{topo} preserves meaningful global trajectory structure.

3 Experimental Design

3.1 Dataset & Modalities

We evaluate our framework on the benchmark **DAIC-WOZ** dataset. The multimodal input space incorporates:

- Audio features
- Facial Action Units (AUs)
- 2D/3D Facial Landmarks
- Gaze vectors
- Head Pose metrics

3.2 Baseline Comparisons

The proposed model is compared against several standard configuration paradigms:

1. Early Fusion
2. Transformer Fusion
3. GATA-Dep
4. **DynDep** (Ours)
5. **GATA-Dep + DynDep** (Plug-in variation)

3.3 Evaluation Metrics & Analyses

Performance is verified using participant-level **F1-score**, **Precision**, **Recall**, and **ROC-AUC**. Additional qualitative and diagnostic analyses include the generation of:

- Recurrence plots
- Persistence diagrams
- Topological biomarker distribution checks
- 2D/3D Trajectory visualizations

4 Expected Impact

The proposed framework shifts the paradigm of automated depression detection from static classification toward dynamical behavioral modeling. Beyond purely maximizing predictive performance, it provides highly interpretable biomarkers describing how behavioral states evolve during clinical interviews, offering a critical bridge between machine learning representations and established psychological theory.